

Is soil texture a major controlling factor of root:shoot ratio in cereals?

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Summary

Estimation of inputs of belowground carbon constitutes a major uncertainty in carbon (C) balance models. Fixed allocation coefficients are widely used although root:shoot ratio is known to respond to abiotic stressors. This pot experiment investigated the C allocation response of spring barley (*Hordeum vulgare*, L.) to soil texture and nutrient availability. Differences in nutrient availability, induced by a long-term history of contrasting N and PK fertilizer application to the soil, did not affect root:shoot ratio. Soil texture had a significant effect; the smallest root:shoot ratio (0.10) occurred in a clay loam soil and the largest (0.22) was in a sandy soil. We hypothesized that this relation was related to hydraulic properties of the root–soil contact zone and were able to confirm this relation with existing datasets from the literature. The results obtained in this study could be useful for future estimates of root-derived C inputs, for example in carbon turnover models, but thorough *in situ* validation with different plant species is required.

Highlights

- We tested soil texture and nutrient controls on root:shoot ratios of spring barley in a pot experiment.
- Soil texture rather than nutrient availability was the most important factor affecting root growth
- The largest root:shoot ratio in the sandy soil might be related to water availability
- The texture effect proved to be significant in a meta-analysis of data from literature

Introduction

Root biomass is an important substrate for the formation of soil organic matter. It has been shown that the stabilization efficiency of root-derived carbon (C) in soil is two to three times greater than that of aboveground crop residues (Kätterer *et al.*, 2011). This is attributable to several root-specific properties, which have been summarized by Rasse *et al.* (2005). Nevertheless, exact quantification of belowground inputs of carbon to soil is difficult and complex. Estimation of root-derived carbon inputs by yield-based general plant allocation coefficients (Bolinder *et al.*, 2007, 2015; Gan *et al.*, 2009) is usual practice, but it constitutes a major unquantified uncertainty in soil carbon turnover models. Current understanding of the mechanisms by which biotic and abiotic factors control plant C allocation under field conditions is rather poor. Therefore, these factors are not addressed adequately when belowground C inputs are estimated. In addition, species-specific genetic control of root architecture (Warren *et al.*, 2015), nutrients, especially nitrogen

(N), phosphorus (P) and potassium (K), salt and water stress are known to affect root growth and thus change plant root:shoot ratios (Fredeen *et al.*, 1989; Ericsson, 1995; Munns, 2002; Wakeel *et al.*, 2005). Furthermore, soil bulk density and shear strength, which are mechanical constraints, have been found to affect root growth considerably (Taylor *et al.*, 1966; Jones, 1983). In agricultural soil, all of these factors that potentially affect root growth and alter internal plant C allocation are triggered by two major site characteristics: (i) soil texture and (ii) nutrient availability. This study examined whether soil texture or nutrient availability, as influenced by long-term contrasting fertilizer treatments, is a better predictor of root:shoot ratio in spring barley, which is one of the major crops in northern European agriculture.

Materials and methods

Soil and study sites

A pot experiment with a malt cultivar of spring barley (*Hordeum vulgare* L. ‘Tipple’) was carried out to investigate and compare the effects of soil texture and nutrient availability on root:shoot ratios

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in a glasshouse under climatic conditions comparable to those in southern Sweden. We chose a pot experiment to restrict the variation in environmental conditions to the two factors of interest. The soil samples used were from three long-term fertilization experiments with almost identical treatments (Carlgren & Mattsson, 2001). The three selected sites were in central Sweden (Bjertorp, 58°23'N 13°13'E and Högåsa, 58°50'N 15°45'E) and southern Sweden (Ekebo, 55°98'N 12°87'E) and their soils differed greatly in their textures. The soil at Högåsa (HO) developed from a sandy glacial outwash (8% clay, 16% silt and 78% sand, sandy loam texture) is classified according to the USDA classification (<https://www.nrcs.usda.gov/wps/portal/nrcs/main/soils/survey/class/>) as a Humic Dystrocrept. The soil at Ekebo (EK) is classified as an Oxyaquic Hapludoll and has a loamy texture (18% clay, 35% silt and 47% sand). At Bjertorp (BJ) the soil is classified as an Aquic Haplocrypt and is a fine-textured silty clay loam (30% clay, 54% silt and 16% sand). Since 1966 (HO and BJ) and 1957 (EK), the soil at each site has been under contrasting fertilizer treatments with four different rates of N application and four different rates of phosphorus and potassium (PK) applications. The N applications were 0, 41, 82 and 125 kg ha⁻¹ at HO and BJ and 0, 50, 100 and 200 kg ha⁻¹ at EK. The PK applications were 0, replacement, replacement +15 kg ha⁻¹ P + 40 kg ha⁻¹ K, and replacement +30 kg ha⁻¹ P + 80 kg ha⁻¹ K at HO and BJ; and 0, replacement, replacement +20 kg ha⁻¹ P + 50 kg ha⁻¹ K, and replacement +30 kg ha⁻¹ P + 80 kg ha⁻¹ K at EK. The replacement treatments represent the amount of PK that is exported through the harvested crop. Those long-term treatments led to pronounced differences in N and available P content (Table 1). Nitrogen was applied as ammonium nitrate, P was applied as mono-superphosphate before 1994 and as triple super-phosphate afterwards, and K was applied as potassium chloride.

The 6-year crop rotation at HO and BJ consisted of barley, oats (*Avena sativa*, L.), oilseed rape (*Brassica napus*, L.), winter wheat (*Triticum aestivum*, L.), oats and winter wheat, and the 4-year crop rotation at EK consisted of barley, oilseed rape, winter wheat and sugar beet (*Beta vulgaris*, L.). The slightly different crop rotation and amounts of fertilizers at EK were because the trials were considered to reflect local agricultural practices. Preceding crops were spring barley at HO and BJ, and oats at EK. The experimental sites are mouldboard ploughed in the autumn and pest management is carried out according to common local agricultural practices. In the present study, the plots used were the unfertilized control (0NPK), all PK amounts without nitrogen (1PK_{0N}, 2PK_{0N}, 3PK_{0N}) and the largest rate of N without PK (3N_{0PK}); this selection reflected an initial focus on PK in the research. The soil was sampled at each of the sites in March 2015 at three random locations in each of two replicate plots of each treatment; the three individual samples were pooled by treatment and site to form a composite sample.

Pot experiment

The fresh, unsieved soil was packed loosely with minimal delay into pots of 13-cm diameter and 15-cm height. We decided to use unsieved but mixed soil as a compromise between keeping

the soil relatively undisturbed (e.g. maintaining soil aggregation) and minimizing the field-scale variation caused by crop residues, macropores or larger stones among replicates. Each fertilizer treatment at each site was replicated five times, giving a total of 75 pots (three sites × five treatments × five pots). During the first days of the experiment the soil settled, especially the fine-textured soil. Nevertheless, bulk density measured at the end of the experiment did not vary greatly between the three soil types (HO, 1.17 g cm⁻³; EK, 1.24 g cm⁻³; BJ, 1.20 g cm⁻³). Four barley seeds per pot were buried 3-cm deep and close to the middle of the pot. Air temperature in the glasshouse was set to 15°C during daytime and 10°C at night to ensure steady, but not too rapid, growth under the nutrient-limited conditions. No extra nutrients were applied during the experiment. The pots were placed on fleece to avoid drought stress and to allow excess water to drain off. The pots and the fleece were watered every 3 days at the beginning of the experiment and daily towards the end (June). The position of the pots was randomized and changed after every watering event, to ensure comparable light and water availability for each plant.

After the seedlings had reached a height of approximately 10 cm, the strongest individual in each pot was kept and the shoots of the other three were cut. Weeds were removed manually after emergence. The plants were harvested when they began to flower, which occurred 70–90 days after sowing. Sampling at the same phenological stage (start of flowering) was considered more important for a comparison of the root:shoot ratio than the total lifespan of the plants. Date of emergence was thus not recorded. However, about 85% of all plants were harvested at day 77 ± 2 after sowing, and we did not observe any consistent treatment effect in the timespan between emergence and flowering date. The aboveground parts of the plants were cut off at the soil surface, dried at 40°C, weighed and crushed. Subsequently, all five aboveground plant samples from each fertilizer treatment and soil (*n* = 15) were pooled, milled, then analysed for total C and N contents with an elemental analyser (LECO-CNS-2000, St. Joseph, MI, USA) and total P by inductively coupled plasma atomic emission spectroscopy (ICP-AES Optima 5300, Waltham, MA, USA) after extraction with HNO₃. Roots were separated carefully from the soil by washing and sieving, dried at 40°C, weighed and ashed overnight at 550°C to distinguish root biomass from adhering soil particles (Misra, 1994). To visualize the spatial distribution of the living root system within the soil, one pot of each soil was scanned by X-ray tomography just before harvesting the respective plants (GE Phoenix vltomelx m, Boston, MA, US).

Statistical analysis

The data (shoot biomass, root biomass and root:shoot ratio) were analysed with a two-way analysis of variance (ANOVA) to test the effect of soil (soil texture), treatment (fertilizer treatment) and their interaction for significance. The statistical distribution of the residuals was assessed with Q–Q plots. After a potential outlier was detected, we performed a χ^2 test, which confirmed this observation. The outlier was removed and the distribution of residuals was then

Table 1 List of sites and treatments, soil pH_{H₂O} values, soil organic carbon, nitrogen and ammonium lactate-extractable phosphorus and potassium contents, soil C:N and C:P_{AL} ratios and shoot C:N and C:P ratios

Site	Treatment	pH	C / %	N / %	P _{AL} / mg 100 g ⁻¹	K _{AL} / mg 100 g ⁻¹	C:N	C:P _{AL}	C:N _{shoot}	C:P _{shoot}
HO	0NPK	6.6	1.7	0.13	2.6	4.0	12.9	638	66	270
HO	1PK _{0N}	6.5	1.8	0.14	3.4	6.8	13.2	528	57	202
HO	2PK _{0N}	6.7	1.9	0.14	9.1	14.9	12.9	205	55	166
HO	3PK _{0N}	6.7	2.0	0.15	10.9	24.0	13.3	179	55	156
HO	3N _{0PK}	6.3	1.9	0.15	2.2	2.7	12.7	869	42	249
EK	0NPK	6.4	2.1	0.14	3.4	7.5	14.9	610	54	204
EK	1PK _{0N}	5.9	2.2	0.14	6.1	11.6	15.8	365	54	191
EK	2PK _{0N}	6.0	2.3	0.16	12.1	18.8	14.4	189	61	178
EK	3PK _{0N}	6.1	1.9	0.13	18.7	29.5	14.2	99	57	198
EK	3N _{0PK}	5.9	2.3	0.16	2.8	7.7	14.1	822	54	233
BJ	0NPK	6.6	1.7	0.15	2.8	11.0	11.3	600	60	339
BJ	1PK _{0N}	6.5	1.7	0.15	3.1	9.0	11.2	543	50	293
BJ	2PK _{0N}	6.5	1.8	0.16	8.3	15.6	11.0	215	54	200
BJ	3PK _{0N}	6.4	1.6	0.15	12.7	25.2	11.1	130	62	159
BJ	3N _{0PK}	6.5	1.9	0.17	1.7	8.1	11.1	1123	42	308

HO, Högåsa; EK, Ekebo; BJ, Bjertorp; C, carbon; N, nitrogen; PAL, ammonium lactate-extractable phosphorus; KAL, ammonium lactate-extractable potassium; C:N, carbon to nitrogen ratio of the soil; C:P_{AL}, carbon to ammonium lactate-extractable phosphorus ratio of the soil; C:N_{shoot}, carbon to nitrogen ratio of the shoot; C:P_{shoot}, carbon to phosphorus ratio of the shoot.

close to normal for all three variables. Homogeneity of variances across groups was analysed with a Levene test. Variances were heterogeneous across the soil types for root biomass and root:shoot ratio. Therefore, we transformed the data to common logarithms (log₁₀), which homogenized the variances across the soil types before the analysis of variance.

The analysis of variance indicated that the effect of soil texture on root:shoot ratios was significant. Therefore, we carried out a literature review using search terms such as soil properties, soil texture and bulk density in combination with root growth or root:shoot ratio so that we could compare our results quantitatively with those in the literature and assess whether the results could be generalized to some extent. A total of six studies was identified that investigated the effect of soil texture on root:shoot ratio of a specific crop. Field and pot experiments were considered and the crops investigated were cotton (*Gossypium spec.*, L.), maize (*Zea mays*, L.), wheat and switchgrass (*Panicum virgatum*, L.). To test the effect of texture for significance across studies, we applied a linear mixed effect model with the continuous variable clay content as a fixed effect and study as a random effect. Again, the residuals were approximately normally distributed. All statistical analyses were carried out in R, version 3.1.2 (R Development Core Team, 2010).

Results and discussion

The fertilizer treatments had no significant effect on root:shoot ratio of spring barley (Table 2). On average, the largest shoot biomass was observed for 3PK_{0N}, followed by 0NPK, 3N_{0PK}, 1PK_{0N} and 2PK_{0N} (Table 3). A similar pattern was observed for root biomass. Both were affected significantly by nutrient availability (Table 2),

Table 2 Summary of the analysis of variance to determine the effect of soil, treatment and their interaction on shoot and root biomass and the root:shoot ratio

Dependent variable	Source	Sum of squares	d.f.	Mean square	F	P
Shoot biomass	Soil	1.486	2	0.743	5.900	0.004
	Treatment	1.894	4	0.474	2.993	0.025
	Soil × treatment	1.774	8	0.222	1.117	0.366
	Residual	8.536	59	0.145		
	Total	11.96	73	1.584		
Root biomass	Soil	3.000	2	1.500	9.584	<0.001
	Treatment	2.835	4	0.708	4.528	0.003
	Soil × treatment	0.808	8	0.101	0.645	0.736
	Residual	9.235	59	0.156		
	Total	15.88	73	2.465		
Root:shoot	Soil	8.819	2	4.410	44.41	<0.001
	Treatment	0.447	4	1.125	1.125	0.353
	Soil × treatment	0.762	8	0.095	0.959	0.476
	Residual	5.859	59	0.099		
	Total	15.89	73	5.729		

d.f., degrees of freedom.

but not consistently across sites (Figure 1). This resulted in a non-significant effect of fertilizer on root:shoot ratio. This is in contrast to previous reports of a change in C allocation in favour of aboveground organs with increasing nutrient supply (Welbank *et al.*, 1974; Hansson *et al.*, 1987). This discrepancy might be attributable to the fact that we did not fertilize the plants during growth, but aimed to investigate the effect of nutrient availability from contrasts in long-term fertilizer applications. Nevertheless,

Table 3 Average shoot and root biomass (g DM plant⁻¹) and root:shoot ratio data with least significant difference (LSD) for the independent variables treatment and site and the model standard error (SE)

Site	Treatment	Shoot / g DM plant ⁻¹	Root / g DM plant ⁻¹	Root:shoot ratio
HO	0NPK	1.04	-0.67 (0.37)	-0.66 (0.22)
HO	1PK _{0N}	0.74	-0.77 (0.38)	-0.66 (0.23)
HO	2PK _{0N}	0.99	-0.82 (0.23)	-0.77 (0.17)
HO	3PK _{0N}	1.32	-0.64 (0.39)	-0.70 (0.20)
HO	3N _{0PK}	0.71	-0.71 (0.33)	-0.58 (0.28)
EK	0NPK	1.28	-0.72 (0.26)	-1.01 (0.10)
EK	1PK _{0N}	0.85	-0.86 (0.19)	-1.03 (0.09)
EK	2PK _{0N}	0.87	-1.05 (0.12)	-1.05 (0.09)
EK	3PK _{0N}	1.38	-0.69 (0.31)	-1.07 (0.09)
EK	3N _{0PK}	1.4	-0.84 (0.26)	-1.05 (0.09)
BJ	0NPK	1.32	-0.85 (0.18)	-0.85 (0.14)
BJ	1PK _{0N}	1.53	-0.87 (0.25)	-0.79 (0.16)
BJ	2PK _{0N}	0.98	-1.06 (0.11)	-0.95 (0.11)
BJ	3PK _{0N}	1.46	-0.93 (0.19)	-0.82 (0.15)
BJ	3N _{0PK}	1.21	-0.97 (0.17)	-0.95 (0.12)
Average	0NPK	1.21	-0.74 (0.27)	-0.84 (0.16)
Average	1PK _{0N}	1.08	-0.83 (0.15)	-0.83 (0.16)
Average	2PK _{0N}	0.95	-0.98 (0.15)	-0.92 (0.13)
Average	3PK _{0N}	1.39	-0.75 (0.3)	-0.87 (0.18)
Average	3N _{0PK}	1.11	-0.84 (0.26)	-0.86 (0.16)
LSD_{Treatment}		0.28	0.29	0.23
HO	Average	0.98	-0.72 (0.21)	-0.67 (0.22)
EK	Average	1.18	-0.83 (0.16)	-0.87 (0.14)
BJ	Average	1.33	-0.94 (0.13)	-1.04 (0.10)
LSD_{Site}		0.21	0.22	0.17
SE		0.14	0.15	0.09

Root and root:shoot data are log-transformed as used in the analysis of variance; back-transformed means are given in brackets.
g DM plant⁻¹, gram dry matter per plant.

the long-term effect of contrasting fertilizer treatments on nutrient availability and uptake was clearly detectable, even without the addition of any fertilizer during the experiment. Plant C:N:P stoichiometry in aboveground biomass changed considerably: on average across sites, the C:P ratio decreased from 270 in unfertilized soil to 170 for the largest amount of PK (3PK_{0N}). Similarly, the C:N ratio with the largest application of N (3N_{0PK}) was much smaller (46) than in the unfertilized soil (60). This indicates that nutrient uptake was therefore affected strongly by fertilizer regime (Table 1), despite the fact that measurements of element contents were carried out on composite samples of each soil. Another explanation for the absence of a nutrient effect on the root:shoot ratio and total biomass production could be the early harvest at the flowering stage. The effect of nutrient availability might become more pronounced when the plant invests most resources in the protein-rich grains, which have a much smaller carbon to nutrient ratio than straw (Mosse, 1990; Rousk & Bååth, 2007).

There was a strong and significant ($P < 0.001$) effect of soil texture on root:shoot ratio (Figure 1, Table 2). This resulted from both a significant increase in shoot biomass and significant decrease in root biomass with the change from coarse (HO) to fine-textured

soil (BJ) (Figure 2, Table 2). Average root:shoot ratio was 0.10 for BJ, 0.14 for EK and 0.22 for HO. The same texture-related trend was observed for all five fertilizer treatments. Although the effect of texture on root:shoot ratio has not been reported directly previously, somewhat similar effects have been identified by other studies (Al-Khafaf *et al.*, 1989; Evers & Parsons, 2003; Li *et al.*, 2005; Khan *et al.*, 2012; Moore & Lawrence, 2013). When the root:shoot ratios reported in these studies were plotted against clay content (Figure 3), a significant ($P = 0.01$) average negative slope of -0.0041 ± 0.0014 (standard error, Table 4) was obtained. A considerable proportion of the observed variability across studies might be related to the different crop species used, which constrains a direct generalization of the results.

A previous review of 59 studies (Bolinder *et al.*, 2007) found an average root:shoot ratio of 0.13 for small-grain cereals, which is similar to the average of 0.16 determined in this study. This suggests (i) a large recovery rate of roots in the present study and (ii) that root growth was not limited by the size of the pots. Bolinder *et al.* (2007) also used a factor of 1.65 to estimate root-derived carbon (including rhizodeposition) from root biomass carbon. Assuming aboveground net primary production of 3 Mg C ha⁻¹ year⁻¹, which is realistic for

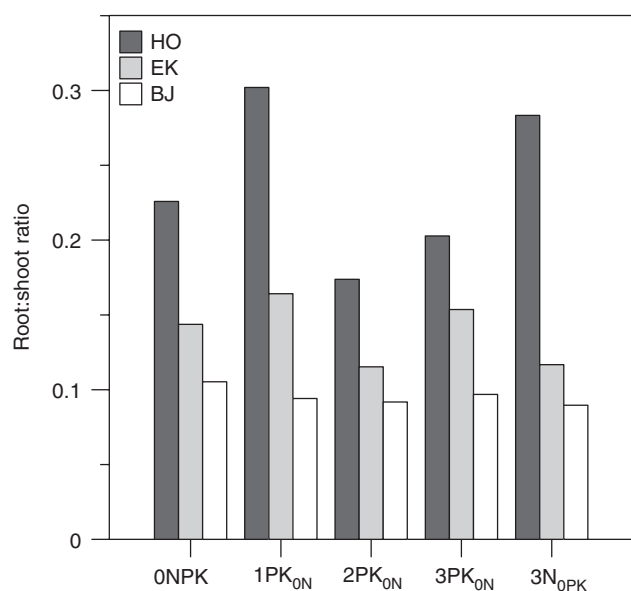


Figure 1 Shoot and root biomass averaged over all treatments for each site (HO, Högåsa; EK, Ekebo; BJ, Bjertorp). The standard error from ANOVA of the $\log_{10}$ -transformed root:shoot ratio values was 0.09 and the respective $\log_{10}$ -transformed average values are given in Table 3. No significant differences between treatments were detected at any site.

cereals grown in a cold temperate climate (Carlgren & Mattsson, 2001), a 10% increase in clay content would decrease the annual root-derived carbon input by $0.12 \text{ Mg ha}^{-1} \text{ year}^{-1}$ when the average coefficient of -0.004 (Table 4) is applied. When root exudates are considered and assumed to change in proportion, which, however, remains unresolved, the annual root-derived carbon input would change by $0.2 \text{ Mg ha}^{-1} \text{ year}^{-1}$. This illustrates that a considerable error might occur when constant values of root:shoot ratio are used to estimate root-derived C inputs.

Several reasons for a soil texture-controlled difference in root:shoot ratio have been demonstrated or suggested: (i) differences in bulk density and thus penetration resistance (Al-Khafaf *et al.*, 1989; Khan *et al.*, 2012), (ii) differences in water-holding capacity and (iii) differences in nutrient availability (Moore & Lawrence, 2013). In the present study, bulk density varied only slightly between soil types and was less than under field conditions ($\sim 1.2 \text{ g cm}^{-3}$). The soil in the pots was watered to excess, which should have limited differences in water availability. Most likely, no water-logging occurred at any time, because the pots were placed on a fleece that allowed excess water to drain. Therefore, oxygen limitation of plants in the fine-textured soil was unlikely. Moreover, differences in the supply and availability of plant nutrients can explain the observed differences in root:shoot ratio to a limited extent only because the effects of long-term fertilizer application were not significant and the C:N ratio of the plant material did not differ between the three soil types.

Plants grown in the clay-rich BJ soil had the largest average biomass and so might have had access to the largest amount of N. However, differences in nutrient uptake were much more

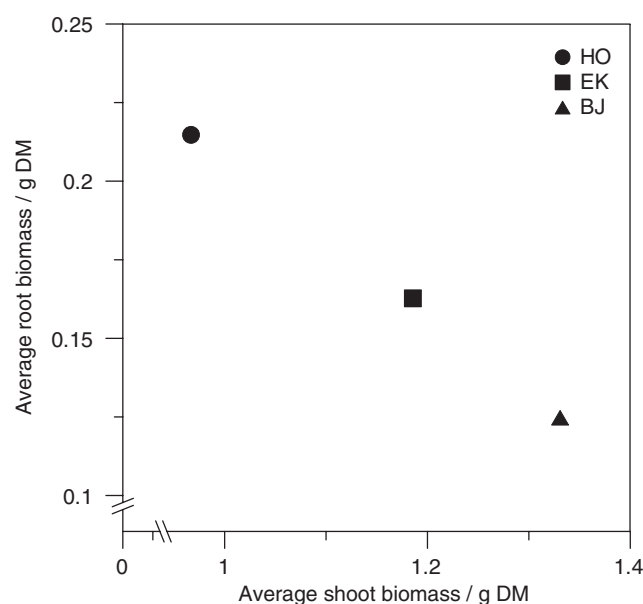


Figure 2 Root:shoot ratio for each treatment and site (HO, Högåsa, 8% clay; EK, Ekebo, 18% clay; BJ, Bjertorp, 30% clay). The standard error from ANOVA of the shoot biomass values was 0.14, and that of the $\log_{10}$ -transformed root biomass values was 0.15; the respective $\log_{10}$ -transformed average values are given in Table 3. DM, dry matter.

Table 4 Summary statistics for the fixed effect of the linear mixed effects model applied to investigate the effect of clay content on root:shoot ratio across a total of seven different studies with estimated coefficients (B), the standard error of the coefficients (SE) and *P* values

Fixed effects	B	SE	<i>P</i>
Intercept	0.3671	0.1111	0.004
Clay content	-0.0040	0.0014	0.013

pronounced between nutrient regimes than between soil types, and nutrient regime did not have any effect on root:shoot ratio. Furthermore, the smallest concentrations of P were in the plants grown on the BJ soil. The most likely explanation for the observed effect of texture might be differences in the direct root–soil interface. Li *et al.* (2005) provided evidence that root–soil contact is related directly to root and root interface hydraulic conductance. Roots in sandy soil are partly exposed to large air-filled pores, which create a physical discontinuity for water movement from soil to root and force the plant to increase the root surface (Herkelrath *et al.*, 1977). Kooistra *et al.* (1992) found that this effect of physical discontinuity was especially pronounced in uncompacted soil, which does apply to the soil used in this study. However, Khan *et al.* (2012) found an effect of texture on root:shoot ratio in compacted soil with a bulk density of up to 1.7 g cm^{-3} . Therefore, the effect probably also occurs under field conditions.

Unfortunately, because of the size of the pots, the resolution of the X-ray scans was too coarse to visualize the roots, which were very fine. However, the X-ray cross-sectional images presented in

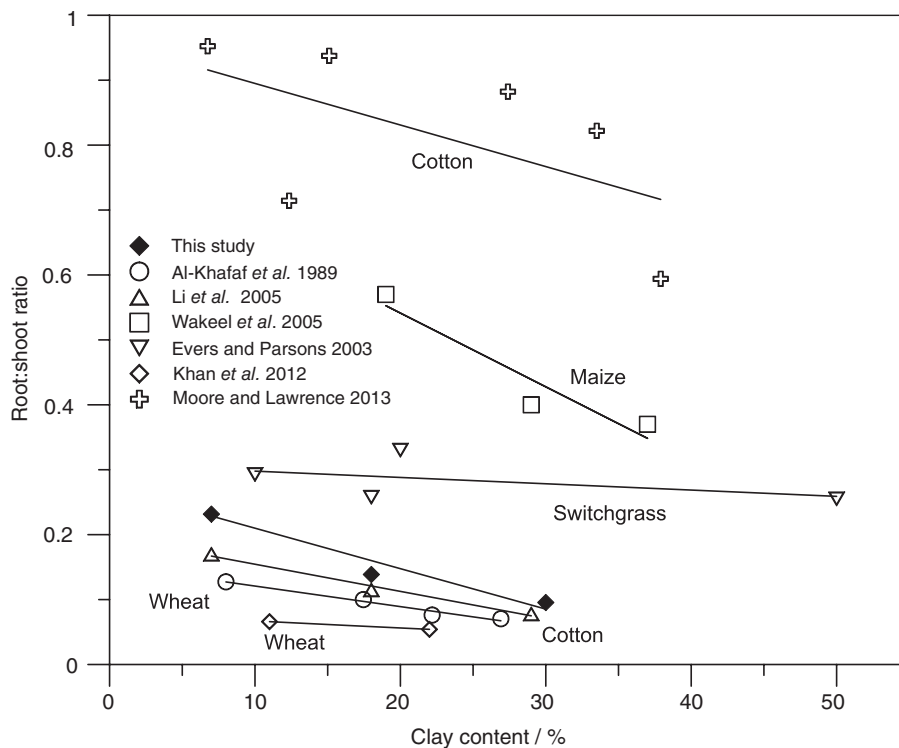


Figure 3 Root:shoot ratios for different crops plotted as a function of clay content (%) for this study and six previous studies; the fitted lines indicate the relations for each individual study.

Figure 4 reveal that spatial variation in bulk density (patchiness) might be more useful for characterizing potential root–soil contact than average bulk density or porosity in soil with different textures. In the loam and clay loam soils of EK and BJ, patches (aggregates) of high density and low porosity were abundant, whereas in the sandy HO soil pores and mineral soil particles were distributed more uniformly. Overall, the observed difference in root:shoot ratio might point to texture-related differences in water availability, despite excess watering. Sandy soil has the smallest water retention (Saxton *et al.*, 1986), and unfortunately watering was not adjusted to the water-holding capacity of the respective soil, but was approximately equal for all pots. Therefore, we cannot distinguish clearly between a direct matrix effect (physical discontinuity) and the effect of water limitation at certain points in time between two watering events. The pronounced and significant texture effects on aboveground biomass (Tables 2, 3) indicate a texture-induced limitation of plant growth. Further research is needed to clarify how this effect is associated with water availability.

Conclusions

Root:shoot ratios of spring barley decreased significantly with increasing clay content or decreasing sand content, whereas long-term fertilizer history, and thus nutrient availability, had no clear effect because of the similar responses of shoots and roots. This effect of soil texture has been observed for several other crop types and might be attributable to less root–soil contact in sandy soil, leading to larger plant C allocation to belowground organs to increase the hydraulic conductance of the root interface. Assuming

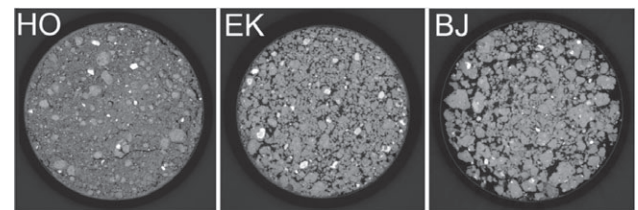


Figure 4 Cross-sections of X-ray scans for each soil (HO, Högåsa, 8% clay; EK, Ekebo, 18% clay; BJ, Bjertorp, 30% clay) at a depth of approximately 2 cm. Black indicates air, white indicates stones and grey indicates material with densities between air and stones.

a fixed root:shoot ratio for any soil might result in a considerable error in the estimation of C input. The results from this research might help to improve estimates of C input, but they need to be validated under field conditions. For this, lysimeter-scale experiments with different soil types at one site could be a suitable option.

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